



Semester: July 2024 –November 2024		
Maximum Marks: 100	Examination: ESE Examination	Duration:3 Hrs.
Programme code: 01 Programme: L.Y.BTECH	Class: LY	Semester:VII (SVU 2020)
Institute/School/Department: K. J. Somaiya School of Engineering	Name of the department: COMP	
Course Code: 116U01E744	Name of the Course: Computer Simulation and modeling	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	List the entities, attributes, activities, events & state variables w.r.to Hospital Emergency room simulation application.	5
ii)	The number of accidents in a year to taxi driver follows Poisson distribution with mean equal to 3. Out of 100 taxi drivers, find approximately the number of drivers with more than 3 accidents in a year.	5
iii)	Explain the event scheduling/time advance algorithm.	5
iv)	Generate 5 random numbers using Linear Congruential Method, use $X_0 = 27$, $a = 17$, $c = 43$, and $m = 100$.	5
v)	What are the types of simulations with respect to output analysis?	5
vi)	What is Empirical Distribution?	5

Que. No.	Question	Max. Marks												
Q2 A	Solve the following	10												
i)	Explain how inter-arrival times are generated in a simulation of a queuing system and why is it essential to ensure that the effective arrival rate is less than the total service rate?	5												
ii)	Describe how randomness is introduced in a queuing system simulation. Why is this randomness crucial for accurately modeling real-world scenarios?	5												
	OR													
Q2 A	With the help of neat diagram, explain different steps in simulation study in detail.	10												
Q 2 B	Solve any One	10												
i)	Consider a Refrigerator selling company. The inventory is maintained as by principle of order-up to-level (M) with review period of N days. The demand is distributed as shown in table.	10												
	<table><tr><th>Demand</th><th>Probability</th></tr><tr><td>0</td><td>0.12</td></tr><tr><td>1</td><td>0.28</td></tr><tr><td>2</td><td>0.15</td></tr><tr><td>3</td><td>0.27</td></tr><tr><td>4</td><td>0.18</td></tr></table>	Demand	Probability	0	0.12	1	0.28	2	0.15	3	0.27	4	0.18	
Demand	Probability													
0	0.12													
1	0.28													
2	0.15													
3	0.27													
4	0.18													

The distribution of lead time is shown as following									
<table border="1"> <tr> <th>Lead Time</th><th>Probability</th></tr> <tr> <td>1</td><td>0.2</td></tr> <tr> <td>2</td><td>0.5</td></tr> <tr> <td>3</td><td>0.3</td></tr> </table>		Lead Time	Probability	1	0.2	2	0.5	3	0.3
Lead Time	Probability								
1	0.2								
2	0.5								
3	0.3								
Assume M = 10 and N = 5 days									
Let simulation start with inventory of 4 refrigerators and order of 6 refrigerators is expected to arrive in 2 days. Generate simulation table for 15 days period consisting of 2 review cycles. Find out how well the current (M, N) system is working with respect to shortage of refrigerators. Random digit for demand: 81, 70, 44, 17, 08, 23, 63, 19, 88, 92, 36, 47, 80, 94, 31 Random digit for lead time: 2, 6									
ii)	Simulate Classic Dump Truck Problem using Event Scheduling Algorithm. Consider following data, Number of trucks: 06 No. of Loaders: 02 No of Scales: 01 Initially two trucks are at loader, one on the scale and three trucks are in loader queue. Consider following activity times <table border="1"> <tr> <td>Loading Times :</td><td>10, 5, 10, 10, 5, 10, 5</td></tr> <tr> <td>Weighing Times:</td><td>12, 16, 12, 12, 16, 12, 12</td></tr> <tr> <td>Travel times:</td><td>40, 60, 40, 80, 100, 40</td></tr> </table> Simulate the dump truck operation and give the system snapshot till time 20	Loading Times :	10, 5, 10, 10, 5, 10, 5	Weighing Times:	12, 16, 12, 12, 16, 12, 12	Travel times:	40, 60, 40, 80, 100, 40		
Loading Times :	10, 5, 10, 10, 5, 10, 5								
Weighing Times:	12, 16, 12, 12, 16, 12, 12								
Travel times:	40, 60, 40, 80, 100, 40								

Que. No.	Question	Max. Marks
Q3	Solve any Two	20
i)	An automatic car wash facility operates with only one bay. Cars arrive according to a Poisson distribution, with a mean of 12 cars per hour. A car may wait in the facility's parking lot if the bay is busy. Service time has a distribution which averages 4 minutes with a standard deviation of $\frac{4}{3}$ minutes. Compute the steady state parameters of the system.	10
ii)	a) A recent survey indicated that 82% of single woman aged 25 years old will be married in their lifetime. Using the binomial distribution, find the probability that two or three woman in a sample of twenty will never be married. b) The hawks are currently winning 0.55 of their games. There are 5 games in the next two weeks. What is the probability that they will win more games than they lose?	10
iii)	Consider the following sequence of 20 numbers, {0.67, 0.31, 0.53, 0.91, 0.80, 0.27, 0.61, 0.49, 0.76, 0.85, 0.62, 0.28, 0.55, 0.77, 0.38, 0.65, 0.29, 0.55, 0.83, 0.92}. Based on the runs up and runs down, determine whether the hypothesis of independence can be rejected, where $\alpha = 0.05$ and critical value $Z_{0.025} = \pm 1.96$	10

Que. No.	Question	Max. Marks																						
Q4	Solve any Two	20																						
i)	Derive the random variate for exponential and uniform distribution using inverse transformation method.	10																						
ii)	Describe in detail the three step approach for model validation?																							
iii)	Consider the following data for past 10 years on demand and lead time. Calculate the Covariance and correlation with respect to multivariate models.	10																						
	<table><tr><td>Lead Time</td><td>5.8</td><td>6.2</td><td>4.3</td><td>6.0</td><td>7.2</td><td>6.5</td><td>5.9</td><td>4.7</td><td>6.8</td><td>6.1</td></tr><tr><td>Demand</td><td>81</td><td>86</td><td>72</td><td>90</td><td>95</td><td>87</td><td>79</td><td>72</td><td>90</td><td>88</td></tr></table>	Lead Time	5.8	6.2	4.3	6.0	7.2	6.5	5.9	4.7	6.8	6.1	Demand	81	86	72	90	95	87	79	72	90	88	
Lead Time	5.8	6.2	4.3	6.0	7.2	6.5	5.9	4.7	6.8	6.1														
Demand	81	86	72	90	95	87	79	72	90	88														

Que. No.	Question	Max. Marks
Q5	Solve any four	20
i)	Point estimation & Confidence interval estimation in output analysis	5
ii)	Goodness of fit tests	5
iii)	Quantile-Quantile Plots	5
iv)	Metamodeling	5
v)	Optimization via Simulation	5
vi)	Verification of Simulation Models	5